AI Network Troubleshooter Report

Filename: advanced_capture2.pcap
Packet Count: 1483

Protocols

TCP: 938
DNS: 73
NTP: 8
ICMP: 20
MDNS: 4
ARP: 6
HTTP: 4
TLSv1: 10
TLSv1.3: 419
BROWSER: 1

AI Diagnostic Report

1. Root Cause Summary:
The packet capture shows predominantly successful TCP communications with a small number (5) of possible failed connections, indicated by the difference between SYN and SYN-ACK packets (17 SYN vs 12 SYN-ACK). Most traffic is HTTPS (port 443) using TLS 1.2, with some legacy TLS 1.0 still present. Key internal IP (172.17.208.219) is the major talker, indicating local host activity. No explicit errors or anomalies were captured.

2. Security Observations:
- Presence of legacy TLS 1.0 in the capture (10 packets) is a potential security risk, as this version is known to have vulnerabilities and is deprecated.
- No suspicious DNS queries or unusual domains were detected. Queries are common well-known domains (google.com, cloudflare.com, openai.com, etc.).
- No indication of scanning or flooding attacks; TCP SYN count low with few failed handshake attempts.
- Use of curl user-agent may indicate scripted or automated requests.
- No explicit evidence of ARP poisoning, DNS spoofing, or ICMP-based reconnaissance was observed.

3. Troubleshooting Recommendations:
- Investigate and resolve the reasons for the 5 possible failed TCP connection attempts to ensure reliable communications.
- Upgrade or disable TLS 1.0 support on clients and servers to enforce stronger security protocols (TLS 1.2+).
- Monitor traffic from the top talker IP (172.17.208.219) to confirm expected behavior, especially regarding any automated curl-based requests.
- Validate firewall and IDS rules for port 443 to ensure no unusual activity is blocked or missed.
- Continue capturing traffic during peak times if connection issues persist, to identify temporal causes.

4. Severity Assessment:
- Low to Moderate severity. The primary concern is the continued presence of TLS 1.0, which is a deprecated protocol posing moderate security risk.
- The small number of failed connections may impact application performance but does not indicate

an ongoing attack or major network failure.
- No critical security incidents such as intrusions or exploit attempts were observed in this capture.